

Assignment 2 – Mini Project

6.3

Customer segmentation
with clustering.

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Contents

Introduction.....	3
Approach.....	3
Analysis.....	4
Data Analysis & Insights.....	4
Clustering with ML models.....	7
Elbow method.....	7
Silhouette Score Method.....	8
Hierarchical clustering.....	9
Cluster Analysis and Interpretation.....	10
Dimension Reduction and Visualization.....	13
Results	15
Conclusion	15

Introduction

This project is focused on customer segmentation. Customer segmentation enables businesses to group customers based on demographics such as age, geography, behaviour amongst other things.

This mini project aims to apply critical thinking and machine learning concepts to design and implement clustering models to perform customer segmentation and improve marketing effectiveness.

Approach

An e-commerce data set containing details about customer, and orders made over a four-year period from 2012 to 2016 was used for this project

The dataset was imported and analysed. There were 951,669 rows, and 20 columns. The data was inspected for missing values and duplicate values. Missing values were identified in city (135), postal code (3716), and State Province (117,192) features, however due to large volume of data these missing values were ignored.

Feature engineering was applied combining or modifying some of the existing features to create new features to help understand the customer base, buying behaviour, preferences and value to the business.

The five features created for the segmentation using feature engineering are:

- Frequency (number of unique orders)
- Recency (time since last order)
- Customer lifetime value (CLV)
- Average Unit Cost
- Customer Age

The data was grouped to create a new dataset *customer_df* with one record per customer.

The resulting dataset was standardised using StandardScaler and machine learning methods used to determine a suitable value of k for the number of customer clusters of segments.

The clusters were visualised using PCA and t-SNE after reducing the hi-dimensional dataset using PCA

Analysis

Data Analysis & Insights

The table below shows an excerpt from the new customer_df dataframe, showing the new features. This shows there are 68,300 unique rows, representing the total number of customers

	Customer ID	Recency	Frequency	CLV	Average_Unit_Cost	Customer_Age
0	1	4404	28	4322.70	89.622857	71.0
1	3	4970	14	2821.00	90.505000	42.0
2	4	4593	8	782.67	37.195556	42.0
3	5	5005	15	2810.98	107.947368	37.0
4	6	4606	13	1604.58	52.526667	66.0
(68300, 6)						

Table 1: Excerpt from customer_df dataframe.

The customer ID column was changed to be the index column.

The descriptive statistics and summary information are shown below

	Customer ID	Recency	Frequency	CLV	Average_Unit_Cost	Customer_Age
count	68300.000000	68300.000000	68300.000000	68300.000000	68300.000000	68300.000000
mean	45447.623397	4625.110864	10.950996	1950.168370	78.894765	51.851625
std	27203.855105	402.904166	8.771057	1719.939245	38.004880	17.477789
min	1.000000	3194.000000	1.000000	0.630000	0.500000	27.000000
25%	22160.500000	4466.000000	5.000000	696.000000	57.579792	36.000000
50%	44140.500000	4766.000000	9.000000	1497.450000	73.449286	52.000000
75%	70423.250000	4940.000000	15.000000	2709.845000	92.515260	67.000000
max	94254.000000	5019.000000	101.000000	18860.960000	1463.500000	82.000000

Table 2: Descriptive statistics for customer_df table

```
RangeIndex: 68300 entries, 0 to 68299
Data columns (total 6 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Customer ID           68300 non-null  int64
1   Recency               68300 non-null  int64
2   Frequency             68300 non-null  int64
3   CLV                   68300 non-null  float64
4   Average_Unit_Cost     68300 non-null  float64
5   Customer_Age          68300 non-null  float64
```

Table 3: Summary information for customer_df

The data was visualised using histograms and boxplots as shown in the figures 1 and 2 below.

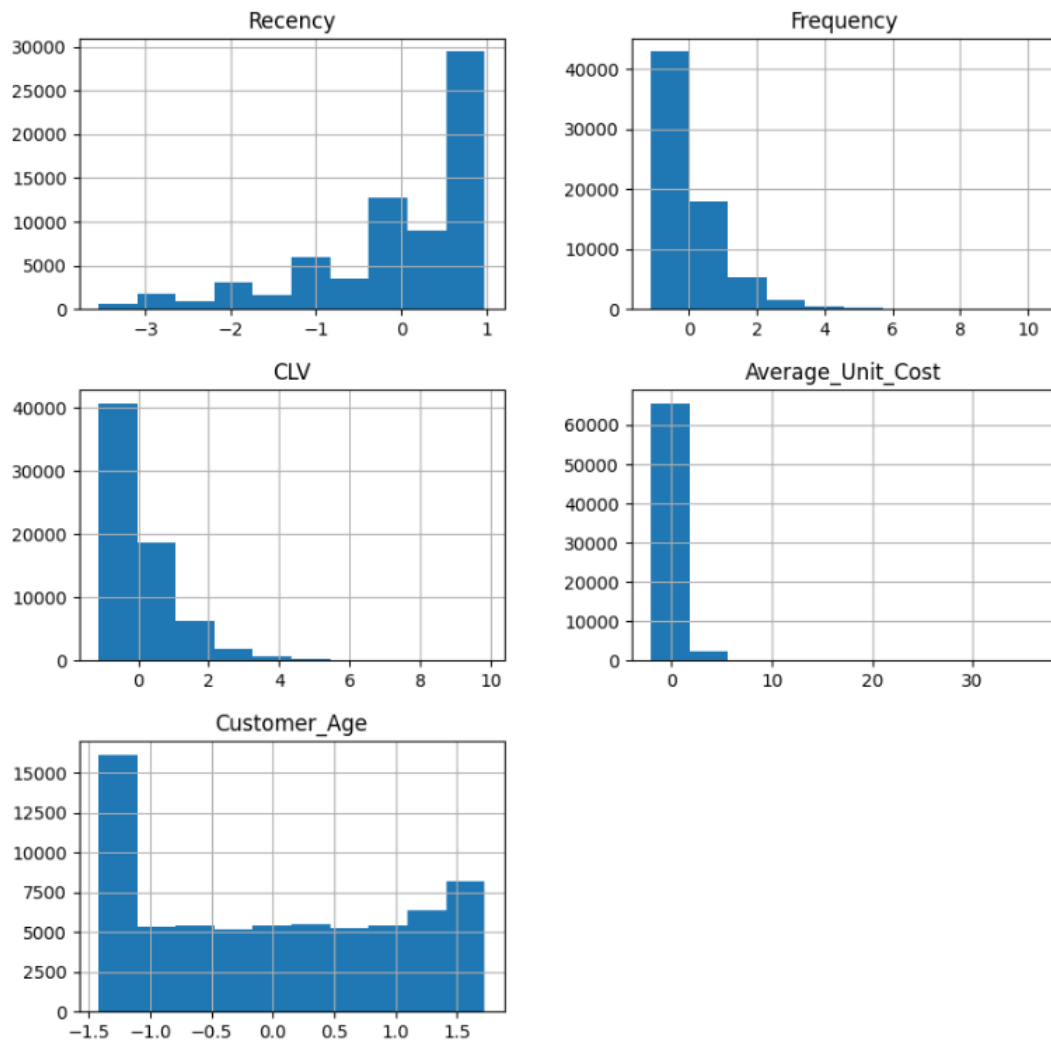
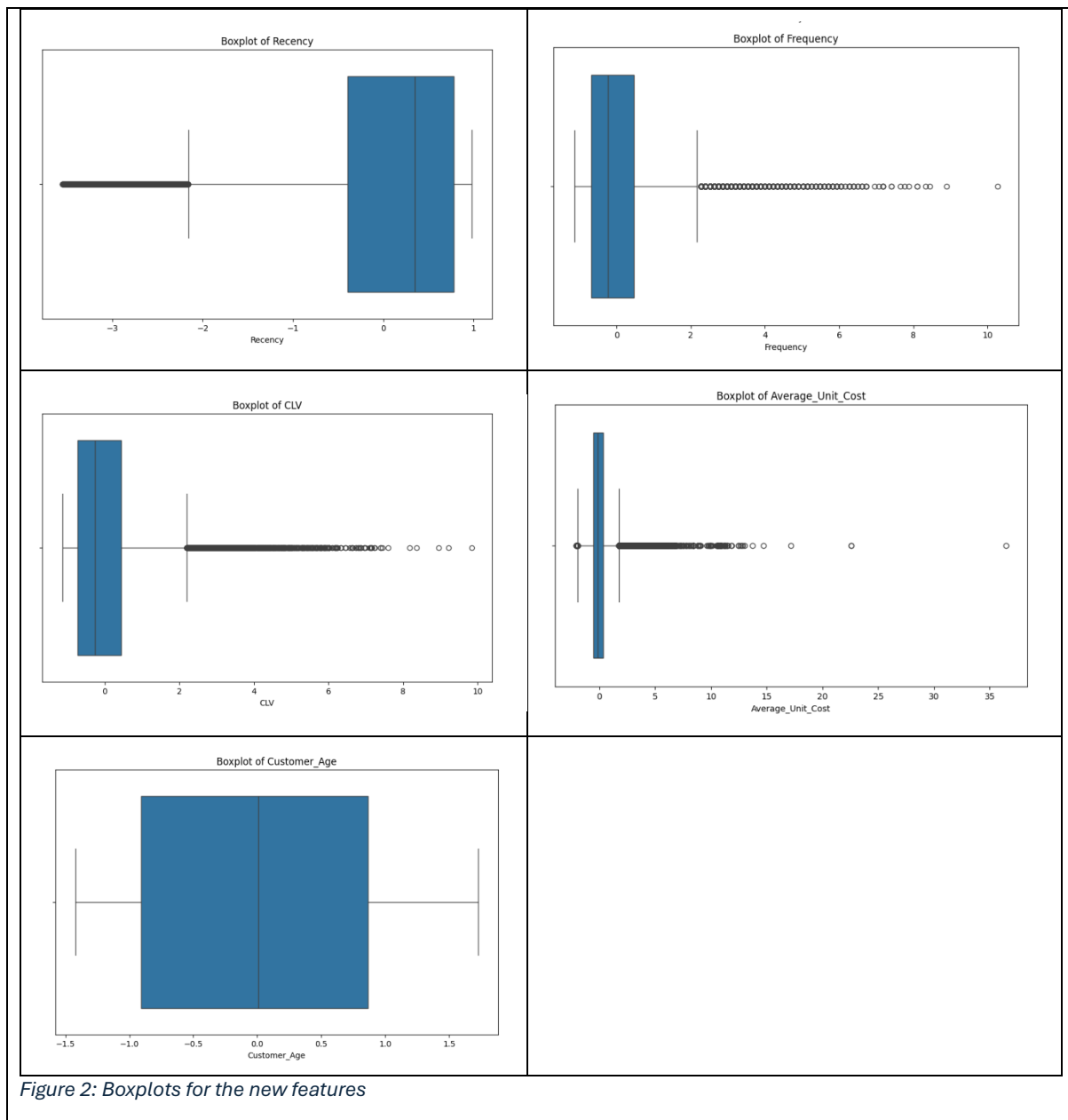


Figure 1: Hist plots for the newly created features.

The figure below shows the boxplots for the newly engineered features



The boxplots show that there are outliers for the features.

The Inter-quartile range (IQR) method was used to identify outliers within the data. The table below shows some of the outliers,

Customer ID	Recency	Frequency	CLV	Average_Unit_Cost	Customer_Age
16	0.774108	0.461636	2.672051	1.493220	1.438887
27	-2.715581	-0.564474	-0.498026	-0.437638	-0.277590
30	-2.536877	-0.678486	-0.230922	1.599067	-1.364692
60	-2.251447	-1.020522	-0.916358	-0.077923	-0.277590
69	-3.298850	-1.020522	-1.118343	-1.798329	-0.849749
71	0.642562	-0.222437	0.481783	1.744185	-0.277590
74	-1.760012	1.601758	3.030659	0.713973	0.523432
77	0.669864	2.627867	3.321945	0.252531	-0.849749
98	-2.683315	-0.792498	-0.723501	0.937524	0.237353
116	-1.263612	-1.134535	-0.854611	1.755188	1.438887

Table 4: Sample of outliers identified using IQR method.

These outliers could indicate unique customer behaviours or data anomalies that might warrant further investigation by the business. A very high CLV outlier could represent a highly valuable customer, while a very low frequency outlier might be a one-time buyer.

Clustering with ML models

K-means clustering was performed using different methods to determine the optimum value of k for the optimum number of clusters.

Elbow method

Different values of k from were tried. The optimum k is where there appears to be an elbow inflection on the line. For this dataset, this appears to be at the 4 for the number of clusters, see figure 3 below.

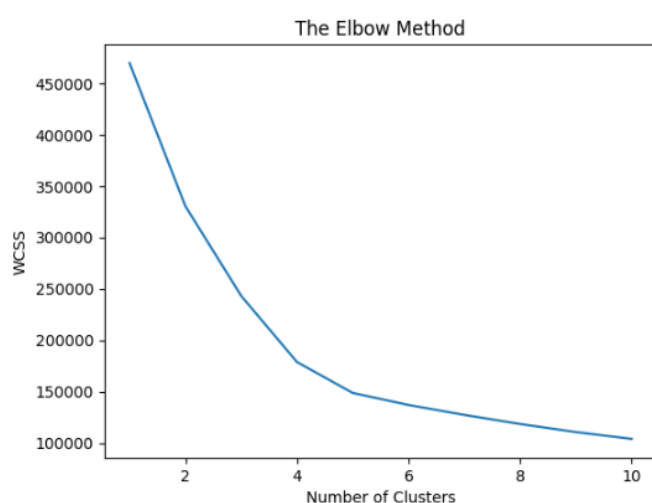


Figure 3: Elbow method

Figure 4 below shows a visualisation of the clusters.

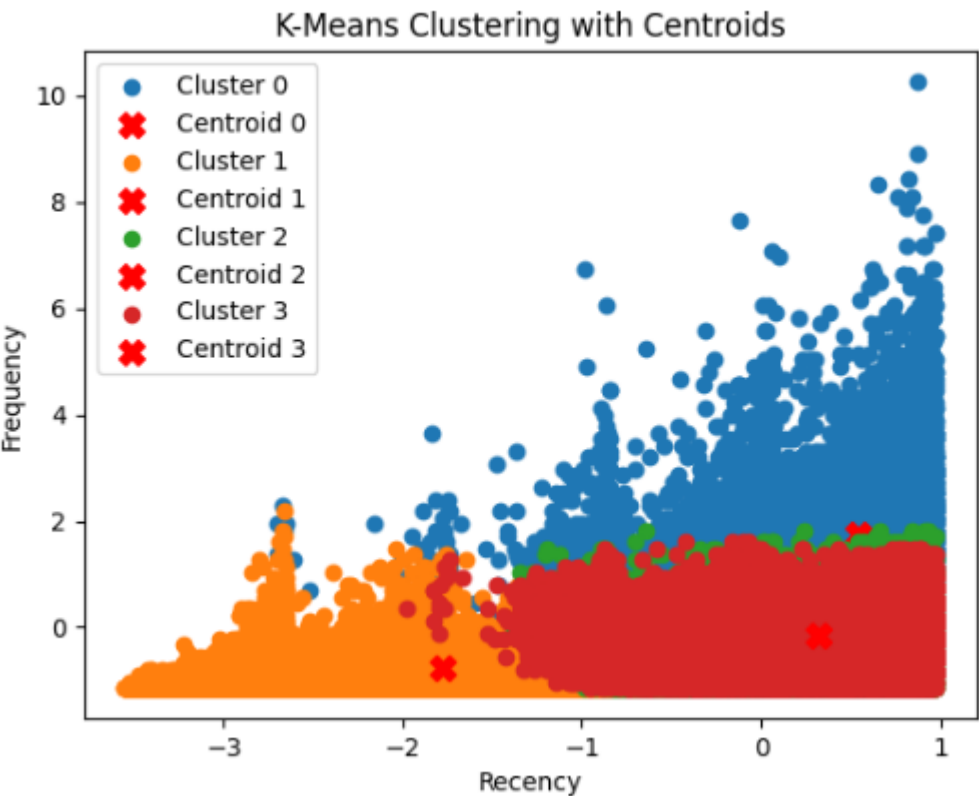


Figure 4: K-means clustering with centroids

Silhouette Score Method

The values of the silhouette scores for different values of clusters were very close, though $n_clusters = 2$ had the highest average score closest to 1. Table 5 below shows the values for the silhouette score and figures 5 and 6 show the visualisations for $n_clusters$ equal to 2 and 5 respectively.

For $n_clusters = 2$	The average silhouette_score is :	0.4091321495572133
For $n_clusters = 3$	The average silhouette_score is :	0.3587311525290901
For $n_clusters = 4$	The average silhouette_score is :	0.33925590299136216
For $n_clusters = 5$	The average silhouette_score is :	0.3528754603442913
For $n_clusters = 6$	The average silhouette_score is :	0.34205230904539347

Table 5: Results for silhouette score

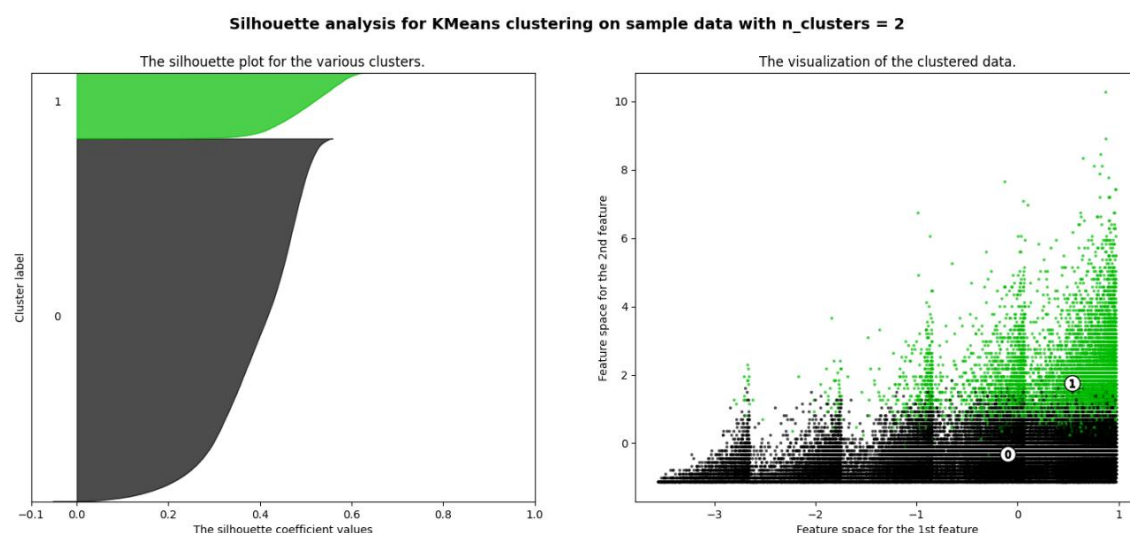


Figure 5: Silhouette plot and Visualization for $n_clusters = 2$

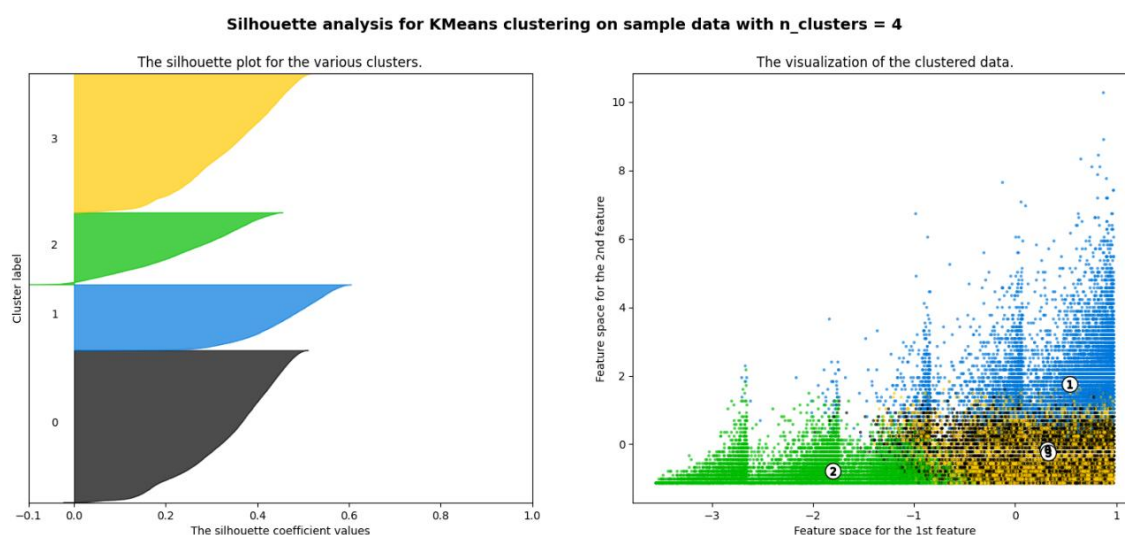


Figure 6: Silhouette plot and visualization for $n_clusters = 4$

From the two methods, the value of $k = 4$ was chosen based on point of inflection on the elbow method.

Hierarchical clustering.

A smaller dataframe `small_df` was created by sampling 30,000 records from the `customer_df` dataframe. This was needed as the notebook was crashing. Hierarchical clustering requires high computational resources due to the complexity of merging or splitting clusters at each step.

The dendrogram is shown in figure 7 below. Cutting just below the 150 mark on the y-axis gives $k = 5$

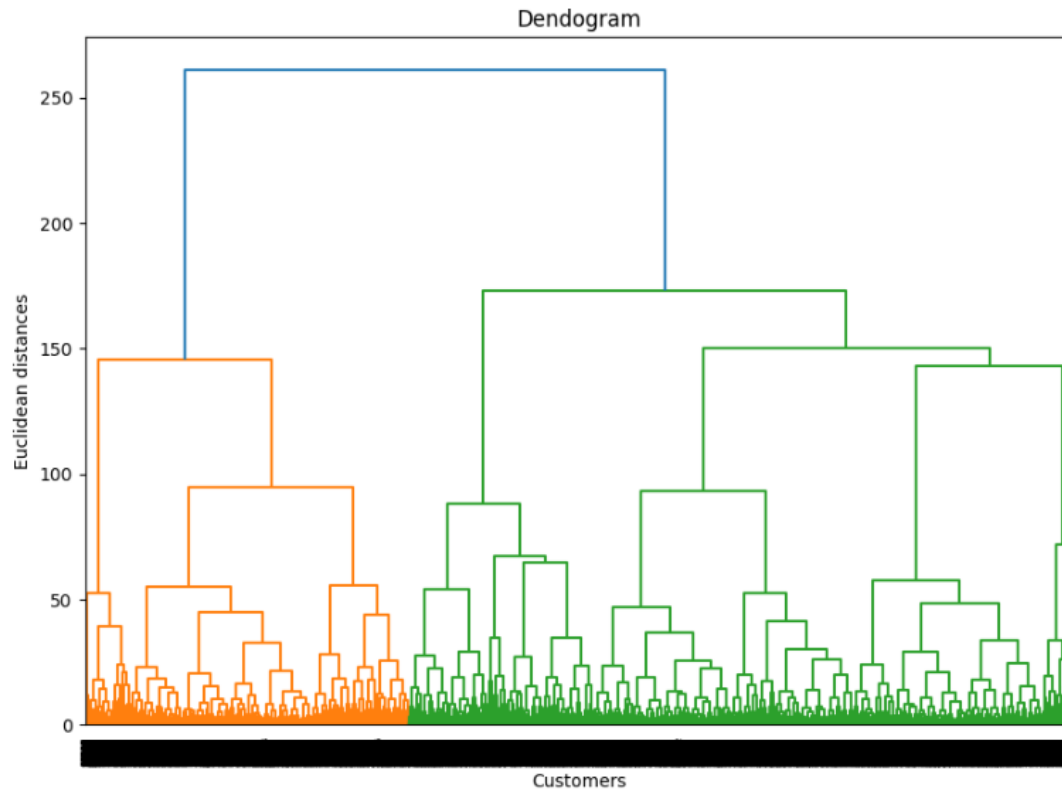


Figure 7: Dendrogram

Based on this data set, each of the techniques can be useful in determining the optimum number of clusters (k), coupled with the business knowledge.

Cluster Analysis and Interpretation.

The table below shows a snapshot of the data with the customer ID and their associated cluster

	Recency	Frequency	CLV	Average_Unit_Cost	Customer_Age	Cluster
Customer ID						
1	-0.548797	1.943794	1.379438	0.282284	1.095591	1
3	0.856014	0.347624	0.506319	0.305496	-0.563670	3
4	-0.079699	-0.336449	-0.678807	-1.097215	-0.563670	3
5	0.942884	0.461636	0.500493	0.764450	-0.849749	3
6	-0.047433	0.233611	-0.200932	-0.693813	0.809512	0

Table 6: Customer ID and associated cluster

A count of the number of customers in each segment is shown in the table below

Cluster #	Number of Customers
3	22338
0	20642
2	13855
1	11465

Table 7: Number of customers per cluster

Boxplots for each of the features for each cluster were generated as shown in figures below.

Clusters with lower recency values represent customers who have made purchases more recently, indicating higher engagement. Conversely, clusters with higher recency values consist of customers who have not made purchases for a longer time, suggesting lower engagement or potential churn. Analysing the median, quartiles, and any outliers within each boxplot can provide a more detailed understanding of the recency behaviour of each customer segment

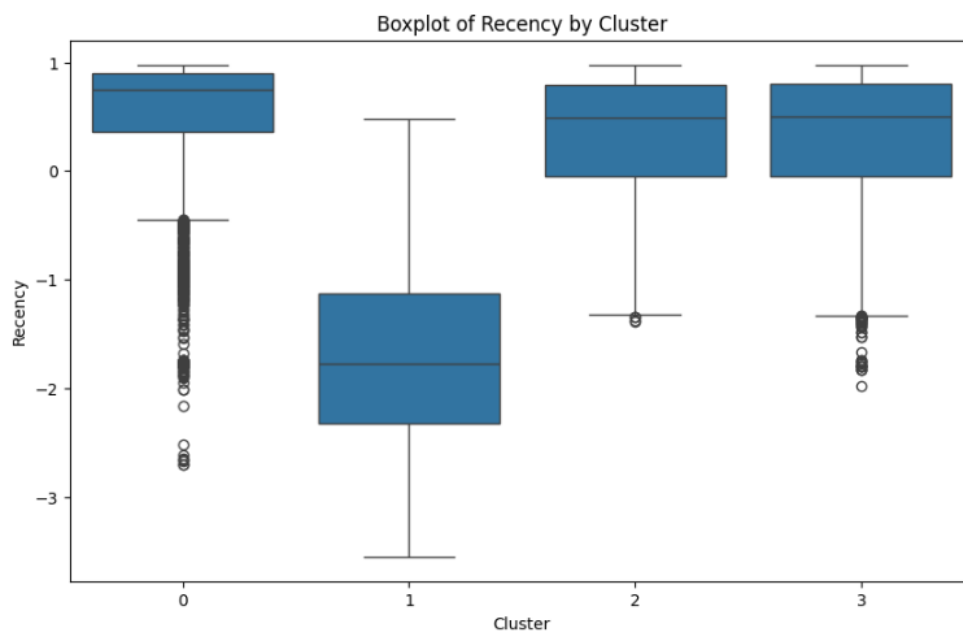


Figure 8: Boxplot of recency by cluster

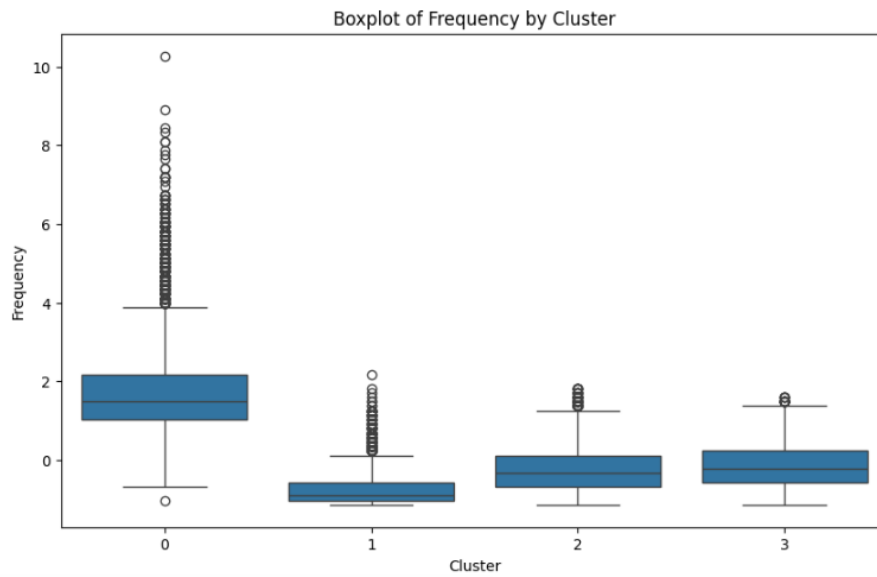


Figure 9: Boxplot of Frequency by Cluster

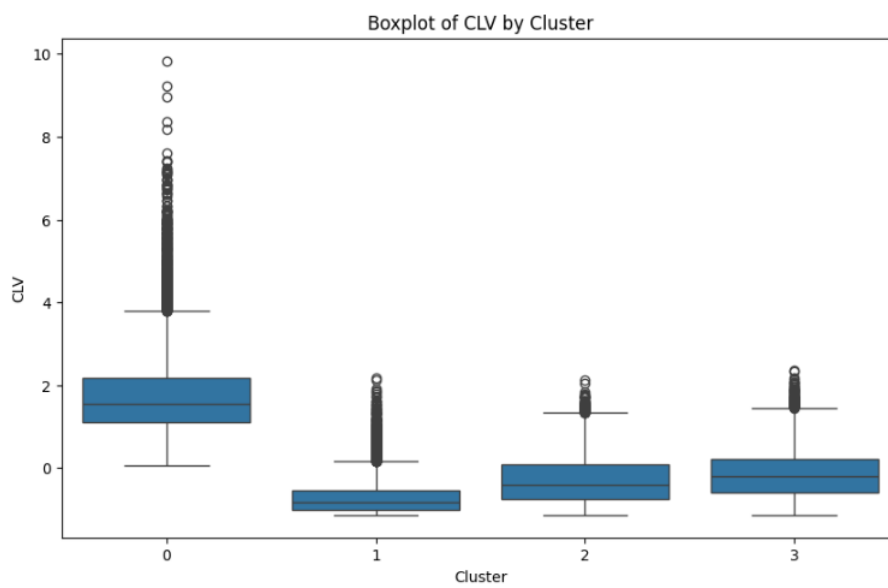


Figure 10: Boxplot of CLV by cluster

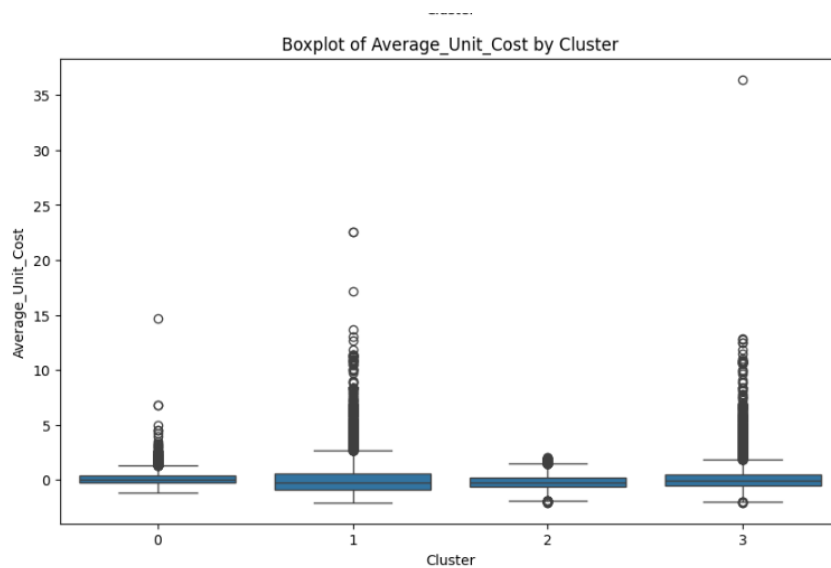


Figure 11: Boxplot of average unit cost by cluster

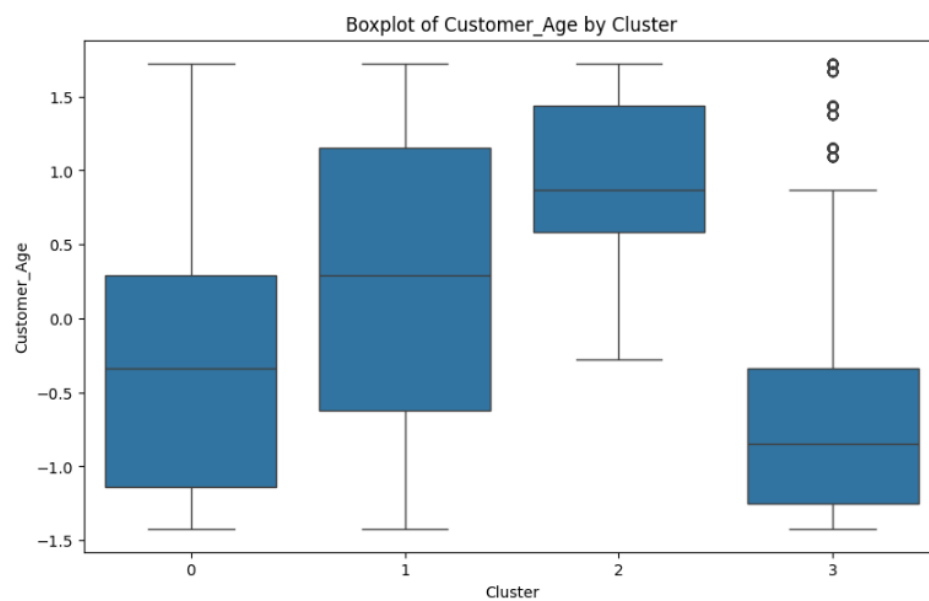


Figure 12: Boxplot of customer age by cluster.

Dimension Reduction and Visualization.

PCA and t-SNE methods were used to reduce the dimensionality of the dataframe and to visualize the dataframe in 2D.

	Recency	Frequency	CLV	Average_Unit_Cost	Customer_Age	Cluster	PCA1	PCA2
Customer ID								
1	-0.548797	1.943794	1.379438	0.282284	1.095591	0	1.690275	0.342154
3	0.856014	0.347624	0.506319	0.305496	-0.563670	3	1.015331	0.120695
4	-0.079699	-0.336449	-0.678807	-1.097215	-0.563670	3	-0.657733	-1.065020
5	0.942884	0.461636	0.500493	0.764450	-0.849749	3	1.212221	0.528174
6	-0.047433	0.233611	-0.200932	-0.693813	0.809512	2	-0.213398	-0.722832

Table 8: Table showing PCA components

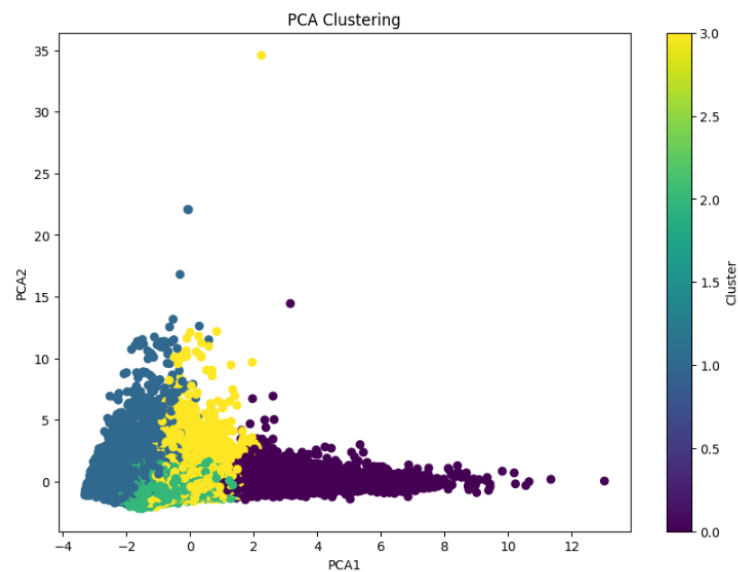


Figure 13: PCA Clustering

The visualization of the t-SNE clustering is shown in figure below.

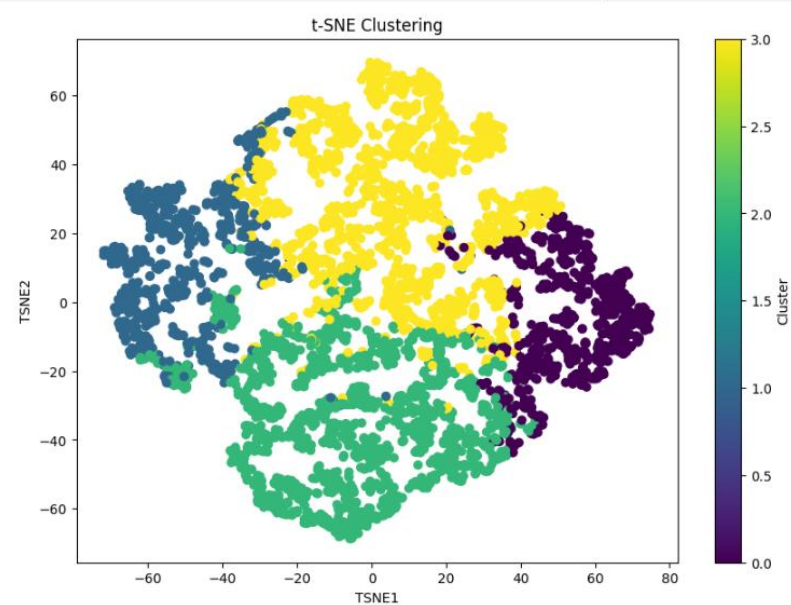


Figure 14: t-SNE clustering

Results

The unsupervised machine learning methods help to select an optimum number of k for clustering. We were able to determine a value of k and identify 4 segments for the customers.

Conclusion

By examining these boxplots with respect to the five features for each cluster, coupled with business knowledge, the company can build a more complete profile of each customer segment, and improve marketing effectiveness. A deeper dive into the data for each cluster can also reveal more characteristics.